Practical No: 1

TITLE

Design a webpage named profile.html that contains the following:
1. A header section with the title "My Profile" and a navigation bar with at least three links.
2. A profile section that includes:
    - o A profile image
    - o Name, short bio, and contact details
3. A skills section displayed using CSS flexbox or grid, containing at least four skill cards.
4. A footer with copyright information.
5. Use CSS3 styling including:
    - o Background color or gradient
    - o Hover effects
    - o Padding, margins, and box-shadow
    - o A media query to adjust layout for mobile screens

OBJECTIVE

To design a responsive webpage named **profile.html** using **HTML5** and **CSS3** that displays personal profile information, skills using Flexbox/Grid layout, navigation links, and a footer, while applying modern CSS styling techniques and media queries.

SYSTEM USED WITH SPECIFICATION / SOFTWARE

Hardware Requirements

- **Computer/Laptop**
- **Processor:** Intel Core i3 / AMD Ryzen 3 or higher
- **RAM:** Minimum 4 GB
- **Hard Disk:** 20 GB free space
- **Display:** 14” or above
- **Keyboard & Mouse**

Software Requirements

- **Operating System:** Windows 10 / Windows 11 / Linux / macOS
- **Web Browser:** Google Chrome / Mozilla Firefox / Microsoft Edge
- **Text Editor / IDE:**
    - o Visual Studio Code
    - o Notepad++

- Sublime Text (any one)

Technologies Used

- **HTML5** – for structuring the webpage
- **CSS3** – for styling, layout, Flexbox/Grid, hover effects, and responsiveness

THEORY

HTML (HyperText Markup Language)

HTML is the standard markup language used to create the structure of web pages. It uses elements such as headings, images, links, sections, and footers to organize content.

CSS (Cascading Style Sheets)

CSS is used to control the presentation and layout of HTML elements. CSS3 introduces advanced features like:

- Flexbox and Grid for layouts
- Media queries for responsive design
- Hover effects and transitions
- Box-shadow and gradients for visual enhancement

Responsive Web Design

Responsive design ensures that a webpage adjusts smoothly to different screen sizes such as desktops, tablets, and mobile devices using **media queries**.

ALGORITHM

1. Start
2. Create an HTML file named profile.html
3. Add a header section with title and navigation links
4. Create a profile section with image, name, bio, and contact details
5. Design a skills section using CSS Flexbox or Grid
6. Add a footer with copyright text
7. Apply CSS3 styling (colors, spacing, shadows, hover effects)
8. Use media query for mobile screen adjustments
9. Save and execute the program in a web browser
10. Stop

PROGRAM

HTML File: profile.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>My Profile</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

<header>
    <h1>My Profile</h1>
    <nav>
        <a href="#">Home</a>
        <a href="#">Skills</a>
        <a href="#">Contact</a>
    </nav>
</header>

<section class="profile">
    <img src="profile.jpg" alt="Profile Image">
    <h2>John Doe</h2>
    <p>Web Developer | Frontend Enthusiast</p>
    <p>Email: john@example.com</p>
</section>

<section class="skills">
    <div class="skill-card">HTML5</div>
```

```
    <div class="skill-card">CSS3</div>
    <div class="skill-card">JavaScript</div>
    <div class="skill-card">React</div>
</section>

<footer>
    <p>© 2026 John Doe. All rights reserved.</p>
</footer>

</body>
</html>
```

CSS File: style.css

```
body {
    font-family: Arial, sans-serif;
    margin: 0;
    background: linear-gradient(to right, #eef2f3, #8e9eab);
}

header {
    background-color: #333;
    color: white;
    padding: 15px;
    text-align: center;
}

nav a {
    color: white;
    margin: 0 15px;
    text-decoration: none;
```

```
}

nav a:hover {
    text-decoration: underline;
}

.profile {
    text-align: center;
    padding: 20px;
}

.profile img {
    width: 150px;
    border-radius: 50%;
    box-shadow: 0 4px 8px rgba(0,0,0,0.3);
}

.skills {
    display: flex;
    justify-content: center;
    gap: 20px;
    padding: 20px;
}

.skill-card {
    background-color: white;
    padding: 20px;
    width: 120px;
    text-align: center;
    box-shadow: 0 4px 6px rgba(0,0,0,0.2);
```

```
    transition: transform 0.3s;
}

.skill-card:hover {
    transform: scale(1.1);
}

footer {
    background-color: #333;
    color: white;
    text-align: center;
    padding: 10px;
}

/* Media Query for Mobile Screens */
@media (max-width: 600px) {
    .skills {
        flex-direction: column;
        align-items: center;
    }
}
```

Output

- A visually appealing **profile webpage**
- Header with navigation bar
- Profile section displaying image and personal details
- Skills displayed in card format using Flexbox
- Responsive layout adapting to mobile screens
- Footer with copyright information

QUESTIONS

1. What is the purpose of CSS Flexbox and Grid?
2. What are media queries?
3. How does hover effect improve user interaction?
4. Why is responsive design important?
5. What is the role of semantic HTML tags?

CONCLUSION

Thus, a responsive personal profile webpage was successfully designed using **HTML5 and CSS3**. The webpage demonstrates the effective use of **Flexbox layout**, **CSS styling**, **hover effects**, and **media queries**, fulfilling all the objectives of the experiment.